

SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem

Bounded source-aligned working unit — 19 July 2026

Authority. Corrected French lines 2573–2581; original printed p. 86; physical source-PDF pp. 76–77; recomposed running pp. 68–69. This bounded unit has been independently source-reviewed against the French TeX and direct PDF; cumulative integration remains pending. Line 2582 is blank; the proof comment begins at line 2583 and is excluded.

Proposition 1.4. — Every left-bounded complex, i.e., every complex for which there exists $q \in \mathbf{Z}$ such that $X^p = 0$ for $p < q$, admits an injective resolution. Moreover, suppose that $u: X^\bullet \rightarrow Y^\bullet$ is a homomorphism of complexes (both left-bounded), and that $x: X^\bullet \rightarrow CX^\bullet$ and $y: Y^\bullet \rightarrow CY^\bullet$ are injective resolutions of $X^\bullet$ and $Y^\bullet$. Then there exists a homomorphism of complexes

$$Cu: CX^\bullet \rightarrow CY^\bullet,$$

unique up to homotopy, such that the diagram

$$\begin{array}{ccc} X^\bullet & \xrightarrow{x} & CX^\bullet \\ u \downarrow & & \downarrow Cu \\ Y^\bullet & \xrightarrow{y} & CY^\bullet \end{array}$$

commutes up to homotopy.